

OrganoidEnv: A Stabilized Reinforcement Learning Environment for Training Biologically Constrained Spiking Neural Networks via Sensory–Motor Mapping

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Summary

This paper introduces OrganoidEnv, a Gymnasium-compatible environment that couples an external deep reinforcement learning (D3QN) controller to a 500-neuron, metabolically constrained Izhikevich spiking neural network (SNN) via current-based sensory–motor stimulation. The SNN incorporates a dopamine-modulated dual-trace eligibility rule, a Global Activity Regulator (GAR), inhibitory homeostasis, and structural pruning to maintain stability and learn from delayed rewards. On a four-stage 2D navigation curriculum, the system reports 70.8% overall success (single run), with ablations suggesting the necessity of the stabilizers and dual-trace plasticity.

Strengths

Technical novelty and innovation

- Framing the SNN as an organoid-like “actor” controlled via stimulation by an external RL agent is an interesting and timely conceptual bridge to organoid intelligence and closed-loop neurostimulation paradigms.
- The integration of metabolic constraints (ATP-like variable), a dual-trace dopamine-inspired plasticity, and stabilizers (GAR, inhibitory scaling, pruning) into a single platform is a noteworthy systems contribution.
- The explicit motor-quadrant stimulation scheme offers a concrete, testable sensory–motor interface for closed-loop control over an SNN.

Experimental rigor and validation

- Inclusion of ablations (dual-trace, GAR/homeostasis, SDM, motor mapping) provides preliminary evidence for the necessity of individual components.
- Baseline comparisons with ANN and RL-only agents situate performance within an upper-bound context.

Clarity of presentation

- The high-level motivation and the positioning relative to organoid intelligence are clearly articulated and accessible.
- Code and logs are made publicly available, supporting reproducibility in principle.

Significance of contributions

- Demonstrates a pathway for stabilizing and shaping recurrent, biologically constrained SNNs using an external RL loop—relevant to neuromorphic computing and AI–neuroscience interfaces.
- Serves as a potential testbed for future hybrid learning rules and hardware studies.

Weaknesses

Technical limitations or concerns

- The paper presents substantive internal inconsistencies in the signal flow: in places the SDM layer “receives a low-dimensional observation vector,” whereas elsewhere the RL agent appears to receive

observations and the SNN only motor stimulation. This ambiguity fundamentally clouds causal interpretation.

- GAR's "global reset" and Poisson-noise rescue are ad hoc controls whose biological realism and algorithmic side-effects are not adequately justified or analyzed.
- The single-seed evaluation for the SNN, coupled with substantial stochasticity, undermines claims of robustness. Reported figures also contain internal contradictions (e.g., Stage 2 success reported as 97% vs. Fig. 2 showing $\approx 5\%$ completion counts in the Stage 2 window).

Experimental gaps or methodological issues

- Core details needed for replication and interpretation are missing or underspecified: Izhikevich parameters (a , b , c , d per population), E/I ratios, connection probabilities/densities, STDP window constants and amplitudes, reward scaling (η , ranges), current-injection amplitudes/durations per action, and per-step SNN simulation window.
- The baselines are not fully aligned for fairness: OrganoidEnv uses D3QN + PER + HER, whereas baselines are labeled as DQN; a D3QN+HER "RL-only" baseline would be the most appropriate upper bound. Conversely, a surrogate-gradient or e-prop SNN baseline would contextualize the choice of dual-trace plasticity.
- No generalization tests (novel goals/obstacle layouts, parameter sweeps), calibration curves, or sensitivity analyses of stabilizer thresholds are provided.

Clarity or presentation issues

- Figures contain placeholders or contradictory axes/scales (e.g., Fig. 4's flat lines at zero; Fig. 2's values inconsistent with text); several typographical and grammatical errors appear throughout.
- The description of the sensory encoding/pathways is inconsistent (diagram vs text), and the sensory-to-SNN mapping is unclear.

Missing related work or comparisons

- Comparison to closely related reward-modulated plasticity frameworks (e.g., DA-BCM-like rules, e-prop-based SNN control, structural-plasticity-on-GPU frameworks) is largely conceptual; direct empirical comparisons are absent.
- The paper would benefit from engaging with recent bio-plausible homeostatic/regularization protocols (e.g., sleep-based stabilization) and newer SNN-RL hybrids addressing delayed credit assignment.

Detailed Comments

Technical soundness evaluation

- The dual-trace eligibility rule is a reasonable three-factor learning choice and is well-grounded in prior literature. However, the lack of specification for plasticity amplitudes, window parameters, and normalization makes it difficult to assess stability and plausibility under different regimes.
- GAR provides coarse-grained global control, but the intervention logic (noise injection vs. resets) could confound learning dynamics by intermittently overwriting internal SNN states. A more principled homeostatic regularization (e.g., intrinsic plasticity, rate targets, or sleep-like consolidation) would deserve consideration and analysis.
- Structural pruning is unidirectional (elimination) and scheduled episodically; the absence of synaptogenesis may exacerbate representational bottlenecks in context-dependent routing. This is acknowledged, but no mitigation is tested.
- Ambiguity around whether the SNN receives exteroceptive inputs (via SDM) or only motor stimulation undermines mechanistic claims about sensorimotor mapping and where learning occurs.

Experimental evaluation assessment

- Using a four-stage curriculum and reporting per-stage performance is appropriate, but single-seed results do not establish reliability. The stark internal inconsistencies between text and figure values (Stage 2) raise concerns about data handling or figure generation.
- The ablation study trends are intuitively plausible (large drops without dual-trace, SDM, or GAR), but single-run ablations and missing error bars prevent firm conclusions.
- The ANN and RL-only baselines vastly outperform the hybrid system, which is understandable given bio-constraints, but the baseline algorithms should match the RL backbone (D3QN+HER/PER) for fairness. Moreover, including a surrogate-gradient SNN and an e-prop SNN would better contextualize the trade-offs.
- Absent are generalization tests (new maps/goals) and stress tests (varying reward sparsity, noise levels, GAR thresholds), which are

important for claiming stability and robustness.

Comparison with related work (using the summaries provided)

- Compared to DA-BCM (Pan et al., 2023), this work adopts a different three-factor rule (dual-trace STDP) and does not co-optimize structure via evolutionary search; however, DA-BCM's empirical baselines (Q-learning, DQN, LSTM) and performance/variance analyses are more comprehensive. A head-to-head experiment with DA-BCM-like rules would be informative.
- Work on GPU-capable structural plasticity (GeNN extensions) demonstrates practical rewiring with negligible overhead. Leveraging such platforms could enable the multi-seed and scaling analyses this work lacks.
- Recent homeostatic regularization via sleep (bio-plausible stabilization of recurrent STDP SNNs) provides theory and systematic evaluation of stability/forgetting trade-offs; integrating or comparing such mechanisms to GAR could strengthen the biological plausibility and stability claims.
- The paper appropriately cites classic R-STDP and three-factor learning rules, as well as neuromorphic hardware (Loihi 2), but omits empirical comparisons to modern SNN training schemes (e-prop, SG-based SNNs) on control tasks.

Discussion of broader impact and significance

- The organoid-intelligence framing is compelling: the external RL agent as an automated experimentalist aligns with real closed-loop organoid paradigms. A robust, open-source simulator that standardizes this setting would be valuable.
- The observed performance gap relative to ANN/RL-only agents is expected but contextualizes the cost of biological constraints. Clearer articulation of when and why one would prefer the constrained SNN (e.g., energy, eventual neuromorphic deployment, interpretability of spiking assemblies) would clarify significance.
- If stabilized and made reproducible, OrganoidEnv could be a useful community benchmark for hybrid SNN-RL approaches exploring realistic constraints.

Reproducibility and reporting

- Public code availability is a major positive. However, critical experimental details (biophysical parameters, stimulation amplitudes,

RL network architecture/optimizer/schedules) must be added to ensure full reproducibility.

- Figures need to be regenerated with consistent scales and accurate legends; contradictions versus text must be resolved.

❓ Questions for Authors

1. Please clarify the full signal flow: does the SDM layer receive exteroceptive observations directly, or are observations only provided to the D3QN agent with the SNN receiving motor-only stimulation? If both occur, how are these streams combined and on what timescales?
2. What are the exact Izhikevich parameters (a, b, c, d) for each population, the E/I ratio and connectivity probabilities, synaptic delays (if any), and the initial weight distributions by connection type?
3. What are the precise STDP window functions, amplitudes, and normalization/decay constants used for the fast/slow traces? How is η chosen and scaled relative to reward magnitudes?
4. How long (in milliseconds) is the SNN simulated after each RL action, and what are the amplitudes/durations of current injections for each motor quadrant/action?
5. How is GAR implemented in practice: what constitutes a “global reset,” what Poisson noise parameters are used, and do these interventions affect weight updates or eligibility traces?
6. Figure inconsistencies: Stage 2 is reported as 97% success, yet Fig. 2 appears to show $\approx 5\%$ completion in the Stage 2 window. Fig. 4 also shows flat lines at zero. Could you reconcile these discrepancies and provide corrected figures?
7. Why do the baselines use DQN while OrganoidEnv uses D3QN with PER and HER? Could you add a D3QN+HER PER “RL-only” baseline and, conversely, a surrogate-gradient or e-prop SNN baseline to isolate the value of your bio-plausible learning?
8. Can you report multi-seed results (even if reduced-episode runs) to estimate variance, and provide sensitivity analyses for GAR thresholds, inhibitory scaling factor α , and pruning thresholds?

9. How does performance generalize to unseen arenas/goals/obstacle configurations? Have you evaluated robustness to sensor noise or reward sparsity?
10. Do motor-quadrant neurons develop stable direction-selective ensembles over training? Can you provide quantitative analyses (e.g., tuning curves, mutual information) supporting the claimed sensorimotor mapping?
11. How often does GAR intervene during successful runs, and does intervention frequency decrease over training? This would help argue that the SNN's stability becomes endogenous.
12. Could you provide energy or event-rate metrics and a preliminary mapping/estimate to neuromorphic hardware (e.g., Loihi 2) to support prospective efficiency claims?

Overall Assessment

The paper addresses an important and timely problem—closing the loop between biologically constrained SNNs and external RL agents under realistic stability and plasticity constraints—and offers an appealing conceptual bridge to organoid intelligence. The systems integration (metabolic constraints, dual-trace plasticity, GAR, inhibitory scaling, pruning) into an open-source environment is promising, and the ablation trends are intuitively consistent with the proposed mechanisms. However, the current manuscript has substantial shortcomings that preclude publication in its present form: key architectural ambiguities (particularly the sensory-to-SNN pathway), missing methodological details needed for replication, single-seed evaluations without variance estimates, and several contradictory or placeholder figures that undermine confidence in the reported results. Baseline choices are also misaligned with the proposed RL backbone, and important comparisons to contemporary SNN learning methods are absent.

I recommend major revisions. Resolving the architectural inconsistencies, fully specifying parameters and stimulation protocols, correcting the figures, adding fair and comprehensive baselines (including D3QN-only and SNN training alternatives), and providing multi-seed/sensitivity analyses would substantially strengthen the technical case and the contribution's value to the Frontiers in Neuroscience community.

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Questions or feedback? Contact us at aireviewer@cs.stanford.edu